

Vishal Kumar Allam

Dallas, TX | 816-666-2813 | itsmevishal36@gmail.com | **Work Authorization: H1B**

PROFESSIONAL SUMMARY

- Data Engineering professional with **5+ years of experience** designing and delivering scalable data architectures, cloud-based data platforms, ETL/ELT pipelines, batch and real-time processing solutions, and analytics workflows using Python, SQL, Spark, Hadoop, Java, and modern cloud technologies.
- Strong hands-on experience with **data gathering, storage, ingestion, transformation, integration, preprocessing, validation, and consumption**, building reliable pipelines that move structured and unstructured data from repositories to analytics and business consumers.
- Experienced with **AWS, Azure, Snowflake, BigQuery, Databricks, dbt, Airflow, Spark, Hadoop, SQL, Python, and Java**, supporting cloud data architecture, data warehouse design, analytics pipelines, and scalable enterprise data management systems.
- Built and optimized **ETL/ELT ingestion pipelines, batch workflows, streaming concepts, data models, and warehouse solutions**, improving performance, reliability, data quality, and accessibility across enterprise data ecosystems.
- Applied **statistical analysis, machine learning techniques, data visualization, dashboards, and reporting** to identify business trends, generate actionable insights, and communicate findings clearly to technical and non-technical stakeholders.
- Collaborated directly with **clients, architects, product teams, and cross-functional engineering teams** to translate business requirements into solution requirements, operational models, use cases, data architecture, and implementation plans in Agile delivery environments.
- Experienced with **GenAI concepts including prompt engineering, RAG, fine-tuning, and generative model evaluation**, along with CI/CD, Linux, data quality, troubleshooting, process optimization, automation, and data-driven decision-making.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Processing:** Python, SQL, Java, Scala, PySpark, Apache Spark, Pandas, Linux, Distributed Processing, Batch Processing, Real-Time Processing, Data Transformation.
- **Data Engineering:** ETL/ELT, Data Ingestion, Data Integration, Data Federation, Data Cleaning, Data Validation, Data Preprocessing, Data Modeling, Data Warehousing, Analytics Pipelines, Structured/Unstructured Data.
- **Cloud & Data Platforms:** AWS, Azure, GCP, Snowflake, BigQuery, Databricks, Hadoop, Azure Data Services, Cloud Storage, Cloud Data Architecture, Data Management Systems.
- **Pipeline & Orchestration:** Apache Airflow, dbt, Spark, Hadoop, Batch Pipelines, Streaming Concepts, Workflow Orchestration, Data Migration, Pipeline Optimization, Performance Tuning, Data Consumers.
- **Analytics & AI:** Statistical Analysis, Machine Learning, Predictive Analytics, Data Visualization, Dashboards, Reporting, GenAI, Prompt Engineering, RAG, Fine-Tuning, Generative Model Evaluation.
- **DevOps & Engineering:** Git, Azure DevOps, Jenkins, CI/CD, Docker, Terraform, Agile/Scrum, Release Management, Monitoring, Logging, Troubleshooting, Production Support, Automation.
- **Consulting & Delivery:** Client Requirements, Business Process Transformation, Process Optimization, Solution Design, Project Management, Stakeholder Communication, Data-Driven Recommendations, Problem Solving, Technical Documentation, Team Collaboration.

EXPERIENCE

Software Engineer AI *PCMI LLC (Full Time)*

Oct 2024 – Present
Dallas, TX, USA

- Designed and implemented scalable **cloud-based data architectures and enterprise data pipelines** using Python, SQL, PySpark, Palantir Foundry, Azure/AWS services, and distributed processing technologies to support insurance analytics and operational decision-making.
- Built and optimized **ETL/ELT ingestion pipelines** for batch and near-real-time processing, extracting data from relational databases, APIs, files, and enterprise repositories and transforming it into curated datasets for downstream analytics consumers.
- Developed reusable data transformation workflows using **Python, Spark, SQL, data modeling, data cleaning, validation, preprocessing, and reconciliation**, improving integrity, accuracy, reliability, and consistency across complex transactional datasets.
- Partnered with **clients, product owners, architects, and source application teams** to translate business requirements into data solution requirements, operational use cases, target-state architectures, process models, and implementation plans.
- Analyzed operational and financial datasets using **SQL, Python, statistical techniques, dashboards, and reporting**, identifying trends and providing actionable insights that supported business process optimization and data-driven decisions.
- Implemented scalable engineering and DevOps practices using **CI/CD, Git, Azure DevOps, Docker, Terraform, monitoring, logging, performance tuning, and production support**, improving pipeline reliability and deployment consistency.
- Applied **GenAI capabilities including prompt engineering, RAG, model evaluation, and AI-assisted analytics** to enterprise data use cases while contributing reusable pipeline patterns, technical documentation, troubleshooting guidance, and process automation improvements.

Environments: Python, SQL, PySpark, Apache Spark, Palantir Foundry, ETL/ELT, Data Ingestion, Data Modeling, Data Validation, AWS, Azure, Snowflake Concepts, Databricks, Hadoop, Airflow, dbt, Git, Azure DevOps, CI/CD, Docker, Terraform, GenAI, RAG, Prompt Engineering, Production Support.

Software Engineer AI *Client: CVS*

Dec 2023 – Sept 2024
Remote, USA

- Designed and developed healthcare **data engineering and cloud analytics solutions** using Python, SQL, Spark, AWS, and distributed processing frameworks to support enterprise reporting, operational insights, and data-driven business workflows.
- Built scalable **ETL/ELT pipelines** that ingested structured and unstructured healthcare data from APIs, relational databases, files, and cloud repositories, transforming raw information into curated datasets for downstream consumers.
- Developed reusable processing workflows using **Python, PySpark, Spark SQL, data cleaning, preprocessing, validation, and data modeling**, improving data quality and reliability across high-volume healthcare datasets.
- Collaborated with **business stakeholders, application teams, architects, and data consumers** to gather requirements, define current and future operational scenarios, translate business needs into technical solutions, and prioritize delivery activities.
- Performed **data analysis, statistical analysis, trend identification, dashboard development, and reporting** to surface actionable insights and communicate operational findings to both technical and non-technical stakeholders.
- Implemented cloud and DevOps practices using **AWS, CI/CD, Git, Docker, monitoring, logging, pipeline optimization, and production support**, improving scalability, performance, and operational resilience across data services.
- Supported AI-enabled data initiatives using **GenAI, prompt engineering, RAG, generative model evaluation, and machine learning concepts**, connecting governed healthcare datasets with emerging analytical and automation capabilities.

Environments: Python, SQL, PySpark, Apache Spark, AWS, ETL/ELT, Data Pipelines, Data Modeling, Data Cleaning, Data Validation, PostgreSQL, MySQL, Airflow Concepts, Snowflake Concepts, Databricks Concepts, Docker, Git, CI/CD, Monitoring, GenAI, RAG, Machine Learning, Analytics.

Software Engineer *Infor*

Sept 2021 – Dec 2022
Hyderabad, India

- Designed and developed enterprise **data pipelines and data management solutions** using Python, SQL, Spark, Java, and cloud-native technologies, supporting analytics, reporting, and operational applications across distributed business systems.
- Built **ETL/ELT ingestion and transformation workflows** to extract data from relational databases, APIs, flat files, and application repositories, standardizing raw data into curated warehouse and analytics-ready datasets.
- Developed distributed processing solutions using **PySpark, Spark SQL, Hadoop, Python, and data modeling**, improving scalability and processing performance for large structured and semi-structured enterprise datasets.
- Implemented data quality processes covering **data cleaning, validation, preprocessing, reconciliation, schema consistency, and exception handling**, improving the integrity and reliability of downstream data products.
- Worked with product managers, architects, analysts, and business teams to **analyze requirements, identify process optimization opportunities, define solution models, and recommend data-driven improvements** aligned with project and business objectives.
- Built analytics outputs using **SQL, Python, dashboards, reports, trend analysis, and statistical techniques**, helping stakeholders interpret complex datasets and make informed operational and business decisions.
- Applied engineering practices using **Git, Jenkins, CI/CD, Linux, Docker, monitoring, troubleshooting, documentation, and Agile delivery**, supporting reliable deployments and maintainable enterprise data solutions.

Environments: Python, SQL, Java, PySpark, Apache Spark, Hadoop, ETL/ELT, Data Warehousing, Data Modeling, PostgreSQL, MySQL, Oracle, Linux, Jenkins, Git, CI/CD, Docker, Analytics Pipelines, Statistical Analysis, Dashboards, Monitoring, Agile/Scrum.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed enterprise **data integration and processing solutions** using Python, SQL, Java, relational databases, and distributed application technologies to support operational reporting and business analytics across client environments.
- Built reusable **ETL workflows and ingestion pipelines** to collect data from databases, files, APIs, and enterprise applications, transforming and loading information into centralized repositories for downstream consumption.
- Implemented Python and SQL workflows for **data cleansing, validation, preprocessing, transformation, reconciliation, and batch processing**, improving accuracy and consistency across recurring enterprise data processes.
- Supported data architecture and warehouse initiatives through **schema design, relational modeling, data integration, storage optimization, and analytics pipeline development**, helping organize data for reporting and business analysis.
- Analyzed business and operational datasets using **SQL, Python, statistical techniques, trend analysis, reports, and visualizations**, helping teams identify patterns and communicate actionable findings to stakeholders.
- Collaborated with architects, analysts, engineers, and client teams to **translate business requirements into technical solutions, troubleshoot data inconsistencies, recommend process improvements, and support Agile project delivery**.
- Applied engineering and operational practices using **Linux, Git, CI/CD, logging, monitoring, troubleshooting, documentation, and production support**, improving reliability and maintainability of enterprise data workflows.

Environments: Python, SQL, Java, Linux, PostgreSQL, MySQL, ETL, Data Integration, Data Cleaning, Data Validation, Data Preprocessing, Batch Processing, Data Warehousing, Data Modeling, Git, CI/CD, Monitoring, Analytics, Reporting, Statistical Analysis, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.